A security company would like to purchase AI-based object detection sw to use security gates Object detection durations:

6.1, 5.3, 5.9, 5.1, 6.4, 4.9, 4.7, 5.6

To use sw in real variance no more than 0.25 sec.

Should the software recommended? Perform appropriate test $\alpha = 0.05$

a) State the null hypothesis to be tested and alternative hypo

H_0:

H_A:

b) Conduct a statistical test of the null hypothesis and state your conclusion $\alpha = 0.05$

Test statistic:

c) Find the critical values

d) Conclusion and interpretation

e) Find the approximate p-value.

In the specs of sw objects are detected 4sec on avg. The quality control supervisor measures 8 durations

6.1 5.3 5.9 5.1 6.4 4.9 4.7 5.6

Perform the hypotheses test whether the mean detection duration 4secs as it is claimed on the specs?

a) State null hypothesis to be tested and the alternative hypothesis

H_0: H_1:

b) Perform the appropriate test of hypothesis for the object detection durations using $\alpha = 0.01$ Calculate the test statistic

c) Find the critical values

d) Conclusion and interpretation

e) Find the approximate p-value